

A02: Recent Advancements and Future Trends in AI

Section 1: Class Notes

1. Real-World AI Uses

- **Edge AI:** This means putting AI chips directly inside devices like cars or smart phones. The data does not have to travel to a distant cloud server. This makes the AI work much faster and safer.
- **Healthcare:** Doctors use AI to read medical images. It helps them find diseases like cancer much earlier. It also helps select the best treatments for patients.
- **Farming:** Smart tractors use AI cameras to see weeds. They spray medicine only on the weeds, not the good crops. This saves money and protects nature.

2. New Technology: AGI and Space

- **AGI (Artificial General Intelligence):** Today's AI can only do one task well, like playing chess. AGI is the future goal. It will be an AI that can learn, think, and solve any problem just like a human.
- **AI Agents:** These are advanced AI systems that can work by themselves. They can plan a multi-step project and complete it without a human helping them at every step.
- **Space Travel:** Space is too far for fast radio signals. Scientists put AI inside space probes so the robots can steer themselves and fix their own errors on Mars or distant stars.

3. Laws and Ethics

- **AI Bias:** Sometimes AI makes unfair choices because it is learned from bad human data. For example, it might favor one group of people over another for jobs. Scientists are trying to fix this.
- **Data Shortage:** AI models need a lot of text and photos to learn. Experts say we are running out of new human data on the internet. Now, scientists are using AI to create fake data to train newer AI.

- **New Laws:** Governments around the world are making strict rules. Companies must now prove that their AI systems are safe, honest, and do not harm the public.
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Section 2: Key Terms

- **Deep Learning:** A type of AI that copies how the human brain works to find patterns in data.
 - **Agentic AI:** AI systems that can take action and complete big tasks on their own.
 - **Synthetic Data:** Fake data made by computers to help train AI when real data runs out.
 - **Edge Computing:** Doing AI calculations locally on a device instead of sending it over the internet.
 - **Neuromorphic Hardware:** Computer chips are built to look and act like real brain cells to save electricity.
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Section 3: Smart Cities and Sustainability

- **Traffic Control:** AI uses street cameras to see traffic jams. It changes traffic lights in real time to stop gridlock.
- **Public Safety:** AI alerts emergency workers instantly if it detects smoke, fires, or car accidents on city roads.
- **Green Energy:** AI tracks electricity uses across a city. It moves solar and wind power to where people need it most to prevent blackouts.
- **Save Water:** Smart sensors monitor city pipes. The AI can find and flag a water leak before millions of gallons are wasted.